GOVERNMENT OF INDIA

Ministry of Chemicals & Fertilizers

Powering Industrial Growth Through Chemical Innovation

Confidential and Proprietory
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SHRI NARENDRA MODI

HONORABLE PRIME MINISTER, INDIA



MESSAGE

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SHRI JAGAT PRAKASH NADDA

HONORABLE MINISTER, MINISTRY OF CHEMICALS AND FERTILIZERS



MESSAGE

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SHRI MUNINDRA KUMAR

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MESSAGE

The chemicals and fertilizers sector serves as a critical foundation for India's industrial development, agricultural productivity, healthcare ecosystem, and manufacturing competitiveness. As India pursues sustainable industrialization and self-reliance, the Ministry of Chemicals and Fertilizers continues to play a vital role in promoting innovation, investment, domestic manufacturing, environmental sustainability, and technological advancement across multiple strategic industries.

The Chamber of Commerce and Industries of India (CCI India) remains committed to supporting the Ministry's vision by strengthening collaboration between government, industry, research institutions, startups, investors, and technology providers. Through policy advocacy, investment promotion, industrial partnerships, innovation facilitation, skill development, and sustainability initiatives, CCI India aims to contribute towards building globally competitive chemical, petrochemical, pharmaceutical, and fertilizer industries.

This Knowledge Paper provides a comprehensive overview of the Ministry's strategic priorities, industrial ecosystem, policy reforms, investment landscape, emerging technologies, and opportunities for public-private collaboration. It explores pathways for enhancing manufacturing capabilities, encouraging green chemistry, improving fertilizer efficiency, strengthening supply chains, advancing specialty chemicals, and promoting circular economy practices.

Designed as a valuable reference for policymakers, industry leaders, investors, researchers, entrepreneurs, financial institutions, and development organizations, the publication seeks to foster meaningful dialogue and collaborative action that enhances industrial competitiveness, attracts investments, promotes innovation, and supports sustainable economic growth.

As India continues to emerge as a global manufacturing and pharmaceutical hub, strategic partnerships will remain central to achieving long-term industrial resilience and global competitiveness. CCI India looks forward to working closely with the Ministry of Chemicals and Fertilizers in realizing this shared vision.

Munindra Kumar
Director General
CCI India

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Preface

India's chemicals and fertilizers sector occupies a foundational position in the national development architecture. As the third-largest producer of chemicals in Asia and among the top ten globally, India's chemical and petrochemical industry generates an annual turnover exceeding USD 220 billion (Source: Ministry of Chemicals and Fertilizers, Annual Report 2023-24). The fertilizer segment underpins the nation's food security by supplying the agricultural backbone with essential nutrients, ensuring that over 146 million farming households (Source: Agricultural Census 2015-16, Ministry of Agriculture) have access to inputs that sustain crop yields across diversified agro-climatic zones.

This knowledge paper has been prepared by CCI India the Chamber of Commerce and Industry of India as a structured strategic advisory document intended to map the Ministry of Chemicals and Fertilizers' mandate, assess sectoral opportunities, identify execution gaps, and propose a comprehensive collaboration framework through which CCI India can serve as a trusted implementation partner, policy facilitator, and ecosystem builder.

The Ministry of Chemicals and Fertilizers administers three departments , the Department of Chemicals and Petrochemicals, the Department of Pharmaceuticals, and the Department of Fertilizers each of which carries distinct policy responsibilities and economic significance. Collectively, these departments interface with sectors that contribute nearly 12% of India's industrial output, generate over 2 million direct jobs, and influence the export competitiveness of more than 50 product categories (Source: Annual Report 2023-24, Ministry of Chemicals and Fertilizers; DPIIT Trade Statistics).

The purpose of this knowledge paper is threefold: first, to present a rigorous, data-backed analysis of the Ministry's sectoral landscape and strategic priorities; second, to identify systemic gaps and untapped opportunities where industry participation can accelerate policy delivery; and third, to present a concrete, actionable collaboration agenda through which CCI India can add measurable strategic value to the Ministry's objectives.

CCI India envisions becoming the Ministry's preferred non-governmental strategic partner one that bridges regulatory intent with industry execution, academic innovation with commercial deployment, and national ambition with global competitiveness. This paper has been prepared for submission to the Hon'ble Union Minister, senior officials, NITI Aayog, industry leaders, development finance institutions, and international strategic partners.

Chamber of Commerce and Industries of India (CCI India)

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About CCI India

CCI India (The Chamber of Commerce and Industry of India) is a premier national business chamber dedicated to fostering economic growth, industry development, entrepreneurship, innovation, and investment promotion across India. As the voice of Indian business and industry, CCI India serves as a bridge between government, industry, startups, academia, investors, and civil society, facilitating meaningful collaboration for nation-building and sustainable development.

Through policy advocacy, business facilitation, research, stakeholder engagement, capacity building, and strategic partnerships, CCI India works to create an enabling environment for enterprises of all sizes. The Chamber actively supports key national initiatives such as Startup India, Make in India, Digital India, Atmanirbhar Bharat, Skill India, and Viksit Bharat 2047 by connecting innovative enterprises with government opportunities and promoting inclusive economic development.

CCI India operates through sector-specific councils, industry forums, knowledge platforms, and state-level engagements that address emerging opportunities and challenges across sectors including manufacturing, agriculture, technology, healthcare, education, infrastructure, sustainability, MSMEs, and startups. The Chamber regularly organizes policy dialogues, investment summits, business delegations, workshops, conferences, and knowledge-sharing initiatives to strengthen India's economic ecosystem.

With a strong commitment to innovation, entrepreneurship, and public-private collaboration, CCI India strives to empower enterprises, facilitate investment, enhance competitiveness, and contribute to India's journey toward becoming a globally competitive and economically
resilient nation. Its role as a knowledge partner and industry catalyst enables ministries, departments, state governments, startups, and businesses to collaborate effectively in achieving national development objectives

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1.Introduction to the Ministry of Chemicals and Fertilizers

1.1 Constitutional and Institutional Mandate

The Ministry of Chemicals and Fertilizers derives its institutional authority from the Allocation of Business Rules, 1961, as amended over successive decades. The Ministry's mandate spans the full policy lifecycle from raw material sourcing and technology licensing, to production regulation, pricing architecture, quality standards, trade facilitation, and environmental compliance across the chemical, petrochemical, pharmaceutical, and fertilizer sectors.

Under the Seventh Schedule of the Constitution of India, industries of national importance including chemicals, pharmaceuticals, and fertilizers fall under the Union List (List I, Entry 52), giving the Central Government exclusive legislative authority over these domains. This Constitutional placement reflects the strategic criticality of these sectors to national security, food sovereignty, public health, and industrial competitiveness.

1.2 Departmental Structure

The Ministry operates through three constitutionally recognised departments, each carrying distinct mandates:

- **Department of Chemicals and Petrochemicals (DCPC):** Responsible for the formulation and implementation of policies for the chemicals and petrochemicals sector. The DCPC oversees PCPIRs, chemical industrial clusters, speciality chemicals promotion, agrochemicals regulation, and India's participation in international chemical conventions. It also administers the Plastic Waste Management framework and interfaces with the Basel, Rotterdam and Stockholm Conventions. The DCPC plays a pivotal role in steering India's transition toward a circular economy in chemicals.
- **Department of Pharmaceuticals (DoP):** Manages the entire policy ecosystem for the pharmaceutical sector including drug pricing under the National Pharmaceutical Pricing Authority (NPPA), pharmaceutical cluster development, the PLI scheme for pharmaceuticals and medical devices, and India's global pharma leadership under the 'Pharmacy of the World' positioning. The DoP coordinates with the Central Drugs Standard Control Organisation (CDSCO) and interfaces with MSME pharmaceutical manufacturers across the country.
- **Department of Fertilizers:** Oversees India's fertilizer production, import, distribution, and subsidy architecture. The Nutrient-Based Subsidy (NBS) scheme, urea pricing policy, feedstock-linked subsidy rationalisation, and direct benefit transfer mechanisms for fertilizer subsidies all fall under this Department. It also oversees the New Investment Policy for urea capacity addition and manages the PM PRANAM scheme aimed at promoting balanced and alternate nutrient use.

1.3 National Significance

The Ministry of Chemicals and Fertilizers occupies a unique intersectoral position in India's development framework. No other ministry simultaneously addresses food security (fertilizers), public health (pharmaceuticals), industrial competitiveness (chemicals), energy transition (green ammonia, biofuels), and environmental sustainability (chemical safety, plastic waste). This multi-dimensional relevance makes the Ministry a critical node in achieving the Sustainable Development Goals (SDGs), realising Viksit Bharat 2047, and maintaining India's strategic autonomy in global supply chains.

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2.Historical Timeline of the Ministry

2.1 Institutional Evolution

The Ministry has evolved through distinct phases of industrial policy, from state-directed self-sufficiency to market-driven liberalisation, and more recently to strategic industrial deepening under the Aatmanirbhar Bharat framework. Each phase has been characterised by specific policy milestones, institutional reforms, and sectoral transformations.

Year	Milestone	Strategic Significance
1947–1960	Post-Independence Industrial Foundation	Establishment of fertilizer plants under the Industrial Policy Resolution 1948. Hindustan Organic Chemicals Ltd (HOCL) and Fertilizer Corporation of India (FCI) established to ensure strategic self-reliance in critical chemicals.
1970	Essential Commodities Act Application	Chemicals and fertilizers brought under price control mechanisms, ensuring affordability for agricultural and industrial users. This created a regulatory paradigm that persisted for decades.
1980s	Green Revolution Support	Massive urea capacity expansion backed by Ministry policy, directly enabling the Green Revolution's yield improvements. India's food grain production doubled between 1968 and 1985, with fertilizer access as a core driver (Source: ICAR, Ministry of Agriculture).
1991	Economic Liberalisation Impact	Delicensing of many chemical industries under the New Industrial Policy. FDI opened in chemicals and pharmaceuticals. This triggered private sector-led growth and India's emergence as a global generic pharmaceuticals hub.
2002	PCPIR Policy	Introduction of the Petroleum, Chemicals and Petrochemicals Investment Region (PCPIR) concept, creating integrated infrastructure for large-scale chemical manufacturing clusters, attracting both domestic and foreign investment.
2005	NBS Scheme Framework	Nutrient-Based Subsidy (NBS) scheme concept developed to rationalise fertilizer subsidies by linking them to nutrient content rather than product-specific pricing, promoting balanced fertilization and reducing fiscal burden.
2013	Pharma Vision 2020	Launch of the Pharma Vision 2020 policy, positioning India as a globally competitive pharma manufacturer with emphasis on innovation, quality, and export diversification. India's pharma exports reached USD 11.6 billion by 2014 (Source: Pharmexcil).
2021	PLI Schemes Launch	Production Linked Incentive (PLI) schemes launched for Pharmaceuticals (INR 15,000 crore), Key Starting Materials/APIs (INR 6,940 crore), and Medical Devices (INR 3,420 crore), marking a decisive shift toward domestic manufacturing deepening (Source: DoP, MoCF).

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